646340

Course Name: MACHINE LEARNING

Course Outcome

CO1- Understand the concepts of Machine Learning and Intelligent agents.
CO2- Understand and learn knowledge representation and reasoning for the problem-solving.
CO3- Apply basic searching techniques for problem-solving.
CO4- Understand and apply learning techniques.
CO5- Apply and utilize AI knowledge for application in the real world.

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Mid Term Examination, Even Semester 2024-25
B.Tech. (CS/Biotech), 3rd Year, VI Semester
BCSE0105; Machine Learning

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the advantages of using k-fold cross-validation over a traditional train-validation-test split in this scenario. How does cross-validation help in model evaluation and selection?	3	2	A	C
2	How does inductive bias influence the performance of models? Explain the role of inductive bias in machine learning, how it affects generalization, and why different models have different biases. How can the right balance of inductive bias improve model performance?	3	1	A	C
3	A retailer analyzes customer purchase data for a predictive model but finds missing values, outliers, categorical data, and varying feature scales. What preprocessing techniques should be applied to improve model performance?	3	2	A	C
4	A financial analyst develops a stock price prediction model but faces high-dimensional data, increasing complexity and overfitting risks. How can Principal Component Analysis (PCA) help? Explain PCA's role in dimensionality reduction, its ability to preserve variance, its impact on model performance, and the potential trade-offs of applying PCA.	3	2	An	C
5	A real estate company uses Linear Regression to predict house prices based on features like size, bedrooms, location, and age. How does Linear Regression work? Explain its assumptions, how it estimates feature relationships, the importance of	3	4	U	P

	coefficients, and challenges like multicollinearity and heteroscedasticity, along with possible solutions.				

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	How does Polynomial Regression differ from Linear Regression in modeling non-linear relationships? Explain how it transforms the input features and fits the model, and discuss the potential risks of overfitting when using higher-degree polynomials. What steps can be taken to avoid overfitting in this case?	5	2	An	C
7	A healthcare company uses Logistic Regression to predict whether a patient will develop diabetes based on features like age, BMI, and family history. The model outputs probabilities for diabetes (1) or no diabetes (0). Explain how Logistic Regression estimates the probability and the significance of the coefficients.	5	3	A	P
8	A retailer uses a k-Nearest Neighbors (k-NN) classifier to predict whether a customer will make a purchase (1 for purchase, 0 for no purchase) based on features like age and spending score. The model uses k=3 neighbors. Given the following data points: • Customer A: (Age: 25, Spending: 80) → Purchase: 1 • Customer B: (Age: 30, Spending: 60) → Purchase: 0 • Customer C: (Age: 28, Spending: 70) → Purchase: 1 What will the prediction be for a new customer with **age = 27** and **spending = 75**?	5	3	A	P